HARBOUR Insurance Company

A Business Analysis Report

(Feasible recommendations for a 24 Hour AI-Assisted Call Services)

By

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14 December 2025

Executive Summary

This report evaluates the feasibility of introducing 24/7 AI-assisted technology at Harbour Insurance Company and addresses declining SLA standards and improving service quality. The data for May 2025 was analyzed using Descriptive Analytics, Hypotheses Testing, and Simulation Modelling using Simul8, to identify performance gaps across contact channels and geographic centers. The findings tell that customer sentiment is evenly distributed across all centers, with Luton being the worst in performance and Edinburgh the best one. The data also highlights that chatbot is among the best-performing channels meeting SLA requirements, but other channels are behind the standards. The CSAT score also falls between 5–6, which shows that customers are not currently happy with the services. So, the simulation model, keeping in mind the requirements, was developed to handle a large volume of customer enquiries via chatbots and webforms and reduce the burden on staff. It will ultimately increase the company's revenue and automate the services. This approach will also improve SLA compliance, reduce dependency on human staff, and support scalable, cost-effective customer service delivery.

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1 Introduction

Harbour Insurance Company is focusing on improving its current business model as it considers the introduction of a new AI-assisted call service. The report is prepared for the senior management team, including the Chief Executive Officer, Service Director, Finance Director, and Marketing Director, following concerns raised regarding service quality, SLA compliance, and future operational efficiencies. The scope includes analysing operational data to perform hypothesis testing and descriptive statistics to identify performance drivers and factors affecting customer experience.

The company has been growing so rapidly that the existing model built in 2015 is no longer able to provide an excellent customer service experience and is breaching Service Level Agreements (SLAs). Based on the exponential needs, a new CEO proposed model is developed that can handle over 90,000 calls per four weeks, with the aim of providing evidence-based recommendations for system redesign and workflow management. This system will be able to resolve enquiries first via 25 chatbots available 24 hours. If they fail to resolve the enquiry, it will then be transferred via a webform, and the remaining ones will be routed to the AI-assisted call services based on the topic of the enquiry.

2 Methodology

A structured methodology to conduct evaluation first began with analyzing the dataset, and using descriptive statistical technique including summary statistics, drawing graphs and extracting meaningful insights for the business. It also involved developing an understanding of sentiment analysis geographically, cast score for each center, average engagement and resolution time and the type of enquiry based on each of the four channels. Up next, I build hypotheses to better understand the nature of the problem i.e. Is SLA being met by call? And if Luton is performing well compared to the other centers? Are sentiments equally shared geographically? Once built I performed z.test, t.test and non-parametric chi-squared tests respectively to know the specific management concerns.

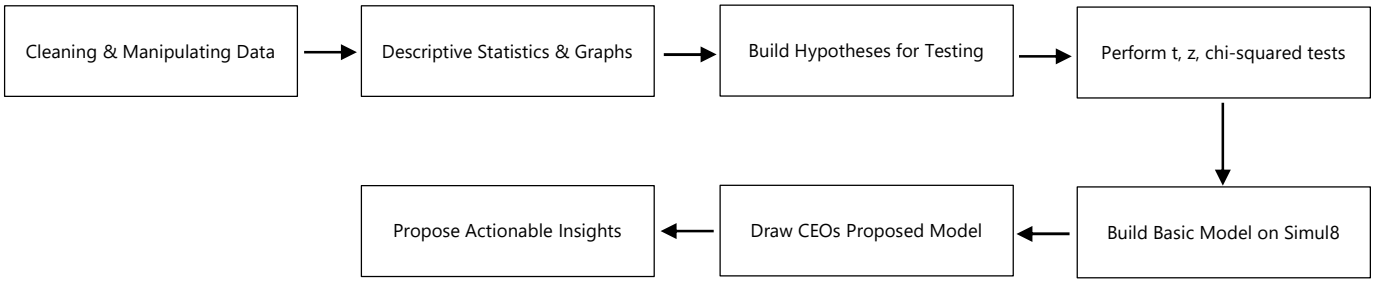

Figure 1: Design Methodology

Finally, simulation modelling using SIMUL8 was employed to represent the dynamic behavior of the service system. Two models were developed; the first one was the basic model to show the current workflow for the company geographically and across different channels. While the second one is the enhanced model to reflect the CEOs propose 24-hour AI-Assisted service center. These two systems were accessed with different parameters of changing staff requirements and number of resources. The output values were used to assess the system capacity, identify bottlenecks, and evaluate the feasibility of meeting future demand targets. Lastly, the project includes proposing actionable business recommendations for the team to improve their customer service and grow the business again.

3 Business Understanding

3.1 Business Objectives

3.1.1 Current Business Situation

Harbour Insurance Company was developed in early 2013 with the focus on travel insurance in Edinburgh. This further expanded its business across the UK, specializing in Home (Edinburgh), Life (Swansea), Travel (Luton), and Vehicle (Liverpool) by maintaining Harbour's core values of transparency and customer service. For now, customers can contact the organization via phone calls, chatbots, webforms, and email services. But the concern raised by the management team was that, based on the negative sentiments and breaking SLA requirements, the company is no longer providing excellent services to its customers. The average CSAT score range between 4–5 shows poor customer satisfaction, and all the centres have a significantly higher number of negative sentiments compared to the positive ones. This also shows that demand volume has increased, and the requirements to govern the standard SLA are no longer being met, as explained in Table 1.

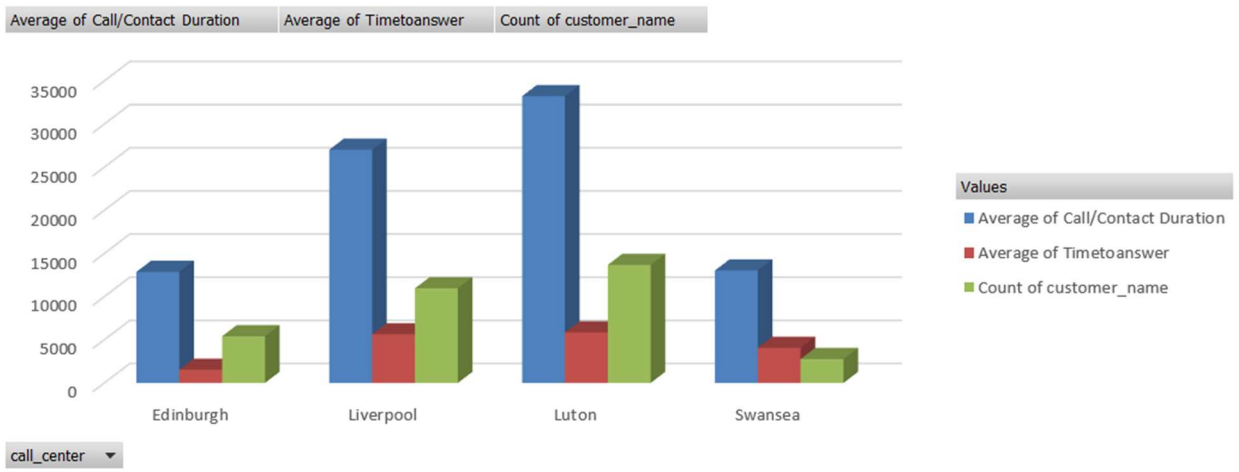

Figure 2: Performance Metrics by Location

Figure 2 shows Luton is the busiest station, with the longest resolution/engagement time and the maximum number of customers. While Edinburgh is the most efficient center despite the fact of handling the moderate number of customers. The data tells that calls are among the most common enquiry types, followed by chatbots. In the same way, billing questions are noticed to be the main concern for most of the customers. Hence, customers can contact the company via any defined channel during working hours from 08:00 AM to 06:30 PM.

3.1.2 Primary Objectives

The primary objective of the company is to improve their customer experience by meeting the SLA requirements and resolving their queries within the dedicated time. This can be achieved not by hiring extra staff to build up the company's budget, but by introducing technological advancements, such as proposing AI-assisted call centres, so customer queries could be resolved faster. It will ultimately affect the performance of the company, increase customer satisfaction scores, and handle a large amount of the growing population effectively. Hence, the company's stakeholders want to redesign their business model so that most of the queries get resolved by the chatbot first, followed by the webform, and those left behind will be assisted by a call identifier by distributing them effectively to the destination centre.

4 Observations with Company Data

The concerns for the senior management team are about introducing a technological system to handle customer enquiries efficiently using language agents. This can be achieved by first knowing the service activities for the company, i.e., to analyze which channel is currently being used most by the customers. The current working activity for the company is that an average duration for resolution and engagement times across all four weeks, which means the performance metrics remain the same regardless of the duration for the month. Also, the right image highlights that the most efficient ways to resolve issues quickly are chatbot and calls having low respond and resolution time. Emails are seen to be the most time-consuming option, having contact duration of more than 24 hours.

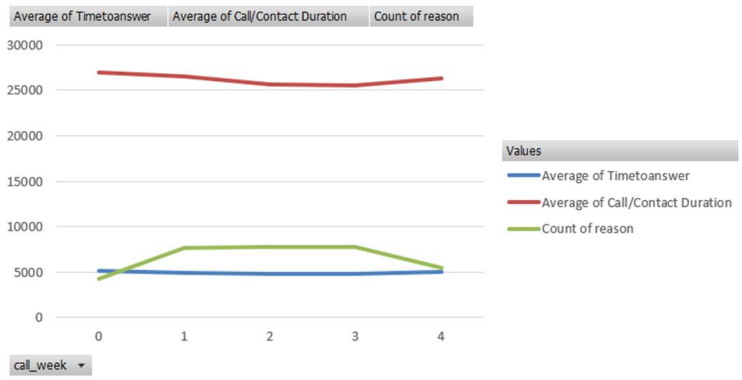

Figure 3: Customer Engagement by Weeks

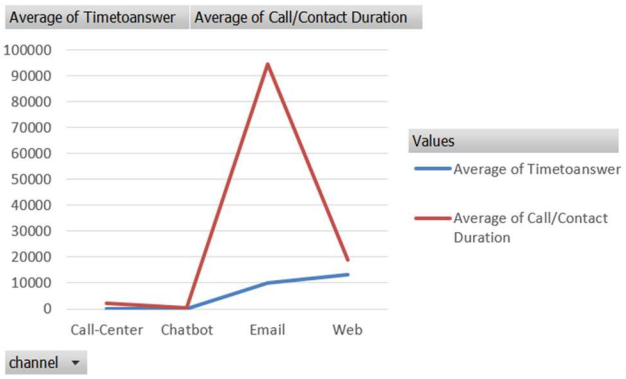

Figure 4: Call Duration by Channel

Figure 4 shows sentiment analysis is distributed equally across all centers with negative ones being the most common and very positive the least, clearly showing many customers still facing unresolved issues across four centers. It also highlights email remains a bottleneck and contradicts with the customer preferences for real-time engagements. Web is used less frequently but still has higher negative remarks, following the same pattern as the other channels. This means there is an opportunity for the company to invest in every contact channel and improve the overall service quality.

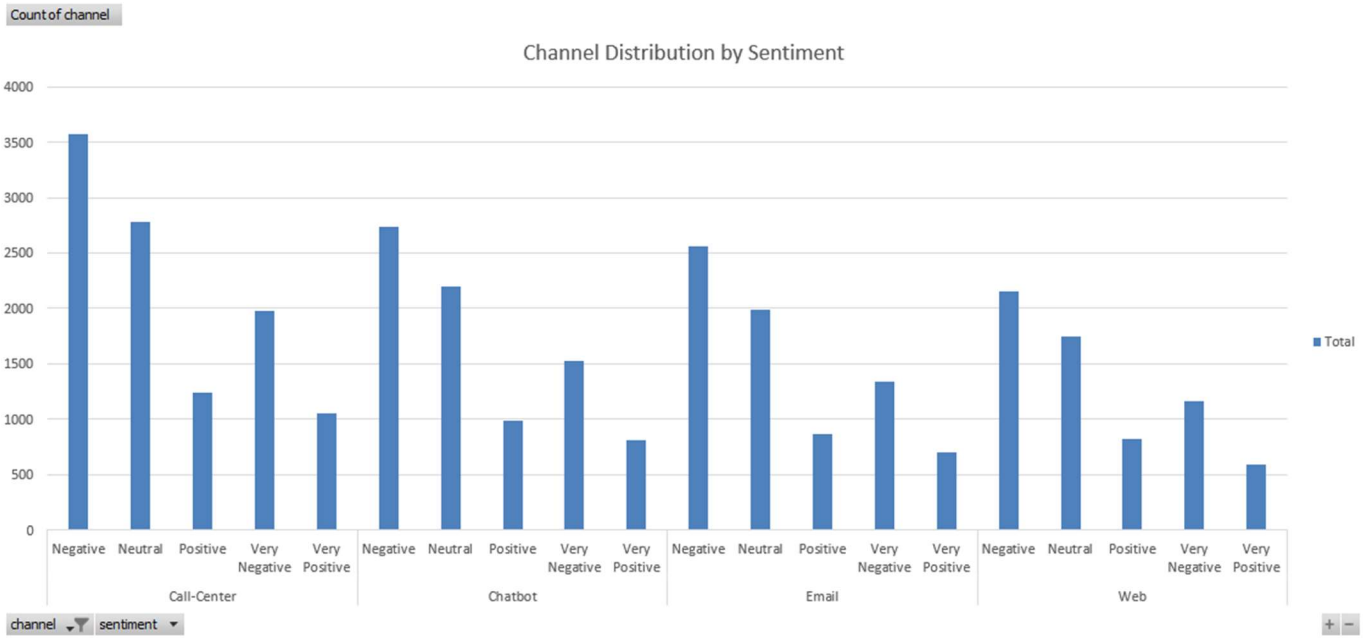

Figure 5: Sentiment Analysis

The table shown below gives us detailed information about the engagement and resolution time for each of the channels. As the SLA for call/chatbot was designed to be 20 seconds, this is being met by the chatbot but exceeded by the call-centre. Also, the chatbot is getting its contact duration SLA met by being under 10 minutes, but not any other channel. On the other side, Email is being the most delayed channel in terms of resolving the customer query. This clearly means that the SLA is being breached and must be redesigned to meet growing customers' needs.

Row Labels	Average of Timetoanswer	Average of Call/Contact Duration
Call-Center	22	2228
Chatbot	14	509
Email	10057	94496
Web	13300	19049
Grand Total	**4922**	**26097**

Table 1: SLA Standards being Exceeded by Channel

The count of reasons raised by the customers during the month of May 2025 is given below in Figure 6. The data highlights that billing questions are seen to be the most common enquiry type, whereas payment questions are seen to be contacted only via the call centre. This can be an opportunity for the business to lessen the burden on human staff by introducing more chatbots specifically designed to resolve payment-related queries. It also warns stakeholders to improve billing services to reduce the burden on the system. Luton has the largest share of customers, whereas Liverpool comes after that. Hence, Travel and Vehicle Insurance are seen to be the top two contact services and must be on the priority list when updating systems. The charts also recommend hiring the most experienced staff in Luton and Liverpool while optimizing the billing services by introducing smooth cash flow and reducing late payments. It also suggests improving self-service options like chatbots and FAQs and redesigning bills to be easily understood.

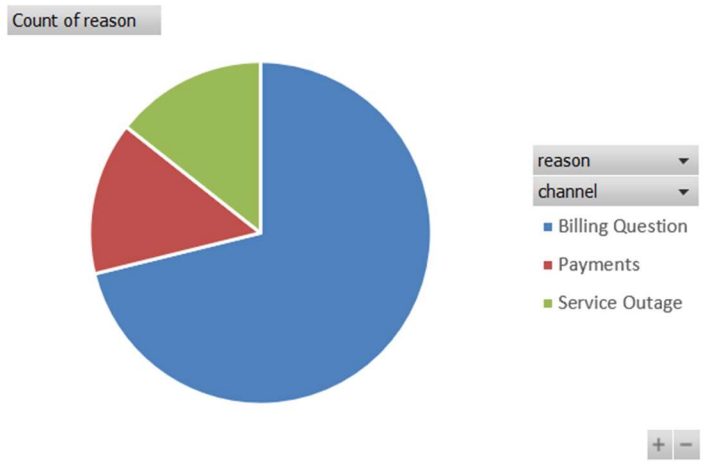

Figure 6: Reasons' Count by Channel

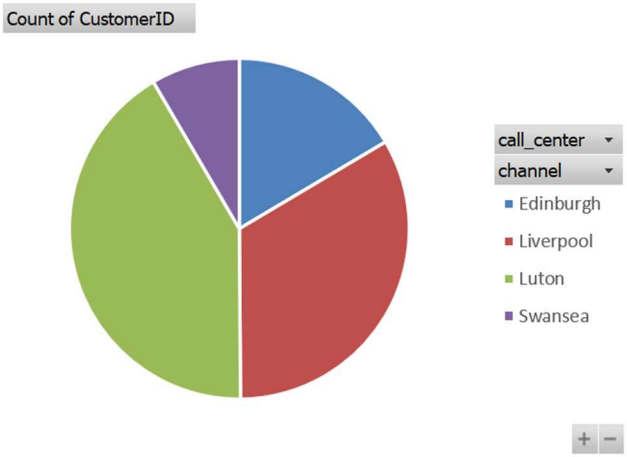

Figure 7: Customer Count by Channel

5 Testing the Data

5.1 Issue 1: Is the busiest channel meeting its SLA standards?

I conducted a z-test to determine whether the defined SLA of 20 seconds is being met when customers contact the company via calls (the most common enquiry channel). The null hypothesis states that the mean time to answer for calls should not exceed 20 seconds. However, after performing the z-test, I obtained a p-value < 0.05 and therefore rejected the null hypothesis. This implies that the SLA engagement time for responding to calls is statistically higher than 20 seconds. Consequently, this represents a serious concern for stakeholders and indicates a need to invest in making call services more effective.

5.2 Issue 2: Are AI-automated channels worth introducing when redesigning the system?

Again, a one-sided z-test was performed to check whether the chatbot is meeting its defined SLA standard of 20 seconds. This test is important for the Service Director's concern that, before introducing additional AI technology, it is necessary to understand the current performance of automated channels, i.e. whether the chatbot is meeting customer needs. The null hypothesis argues that the chatbot meets its engagement requirement of being less than or equal to 20 seconds. Therefore, as the p-value > 0.05, the null hypothesis is accepted reflecting the best performance from chatbots.

5.3 Issue 3: If enquiry resolution time independent of channel type and contact center or not?

To understand if the contact duration is same for calls and chatbot, I conducted a two-tailed t.test. The null hypothesis drawn shows if the mean duration time for call is equal to that of the chatbot; if yes, accept the null hypothesis, otherwise reject it. By performing t.test for both the contact duration columns for calls and chatbots, I came to know the p-value to be less than 0.05, hence rejecting the null hypothesis. It means channel type has a substantial effect on the operational performance. The same way, I performed one-tailed t.test to know if Luton is performing worst compared to other centers based on the high demand of customers or not. The resultant p-value < 0.05, so we would reject the null hypothesis (Luton mean resolution time = other centers mean resolution time). So, the alternative hypothesis validates that Luton's mean resolution time is much higher compared to other centers, a big concern for management team.

5.4 Issue 4: Are customers' sentiment analysis and call reasons evenly distributed among all centers?

This seems a great concern for the management team to know if the customers are experiencing the same service quality across all centres or not. It will be better to manage resources, invest in the right direction, and train the staff based on demand and geography. To understand the problem effectively, I performed non–parametric chi-squared tests on sentiment analysis across all the centres and whether the issue type is independent of geography. The first test conducted on the left tells us the observed and expected tables for sentiments across all centres. The results show that the p-value > 0.05, so we fail to reject the null hypothesis as there is no statistically significant difference that sentiment varies by call centre. Again, the test was conducted with the null hypothesis that call reason is independent of geographic region. The same tables were drawn and formulated that there is no difference in issue types across call centres as the p-value > 0.05.

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6 Recommendations from Statistical Analysis

6.1 Key findings

Almost every center is showing poor customer satisfaction, but I discovered that Luton is worst in response and engagement time while Edinburgh is the best. I also analyzed that chatbot is currently the most effective contact channel whereas calls come after that, showing most of the customers prefer real-time collaboration instead of emails. All payment queries are currently being handled by calls and billing questions. The data also tells service outage does not require human intervention and billing questions are among the most common issue types. The table below highlights these statistics:

Row Labels	Count of reason
Billing Question	**23375**
Call-Center	5890
Chatbot	5901
Email	5901
Web	5683
Payments	**4749**
Call-Center	4749
Service Outage	**4722**
Chatbot	2355
Email	1568
Web	799
Grand Total	**32846**

Table 2: Count of Enquiries

Even though billing questions are higher in count, we still face many delays in the average contact duration. Similarly, SLAs are not being met by contact channels except the chatbot, a big concern for stakeholders. The CSAT score is falling behind the average scale, so it is evident for the company to modernize its services. I also found that the volume for customers remains approximately the same regardless of the start or end of the month.

6.2 Recommendations

Following are the best recommendations for the management team and are included below:

- Prioritize AI-assisted chatbot technology to the system
- Redesign the call center staff and training programs based on demand and geography
- Increased chatbot and webform services as primary reliance for contact
- Update SLA requirements based on current needs

- Invest in AI-tools instead of increasing staff
- Make payment concerns smooth and easy for customers to pay their bills
- Introducing FAQs and feedback forms to better facilitate customer satisfaction
- Route calls automatically based on the type of enquiry
- Analyze specific KPIs that directly impact business growth
- Offer loyalty points to the customers for business reputation

6.3 Suggestions for Further Work

To reduce risks for the future below are some suggestions given:

- Cost benefit analysis for AI-assisted implementation
- Extend simulation scenarios
- Customer sentiment analysis over long times
- Redesign workflow patterns to automate business processes
- Automate billing questions to chatbots instead of calls to reduce burden

7 Simulation of the Call Centre

7.1 Simulation Objectives

The objective of the simulation is to handle over 90,000 calls within four weeks so that business can handle large customers data without service quality delays. The model should work like it handles customer queries via Chatbots, if not resolved then it will be transferred to Webform available 24/7 across all centers. The plan is to automate the tasks and keep human staff requirements as reduced as they can be. This will help LLMs do most of the work and hence smooth the workflow for the system and improve customer satisfaction score.

7.2 Model of the Call Centre

To obtain this I first designed the current workflow for the company showing the queries via Chatbots and Webforms are resolved online whereas email goes to the separate server. For calls there comes human intervention and distribute the calls based on the topics related to Home, Travel, Life and Vehicle etc. This system is an excellent basic drawing to understand the current business model for Harbour Insurance Company and communicate to the stakeholders how this can be improved.

7.2.1 Analysis of Demand

The demand is growing fast, and the current model fails to meet its SLA standards defined in 2015. The system uses 25 chatbots, 10 webforms, and 8 human resources with two shifts across four locations to automate the services. Figure 8 shows that the current model can handle large amounts of customer issues effectively. In the basic model, we can see that most of the issues are resolved by the chatbots and webform and should be prioritized first, as advised in the CEO's model (see Appendix A). This is the logic model that resolves online queries effectively and quickly, likewise reducing dependency on calls.

7.2.2 Current Model of the Call Centre

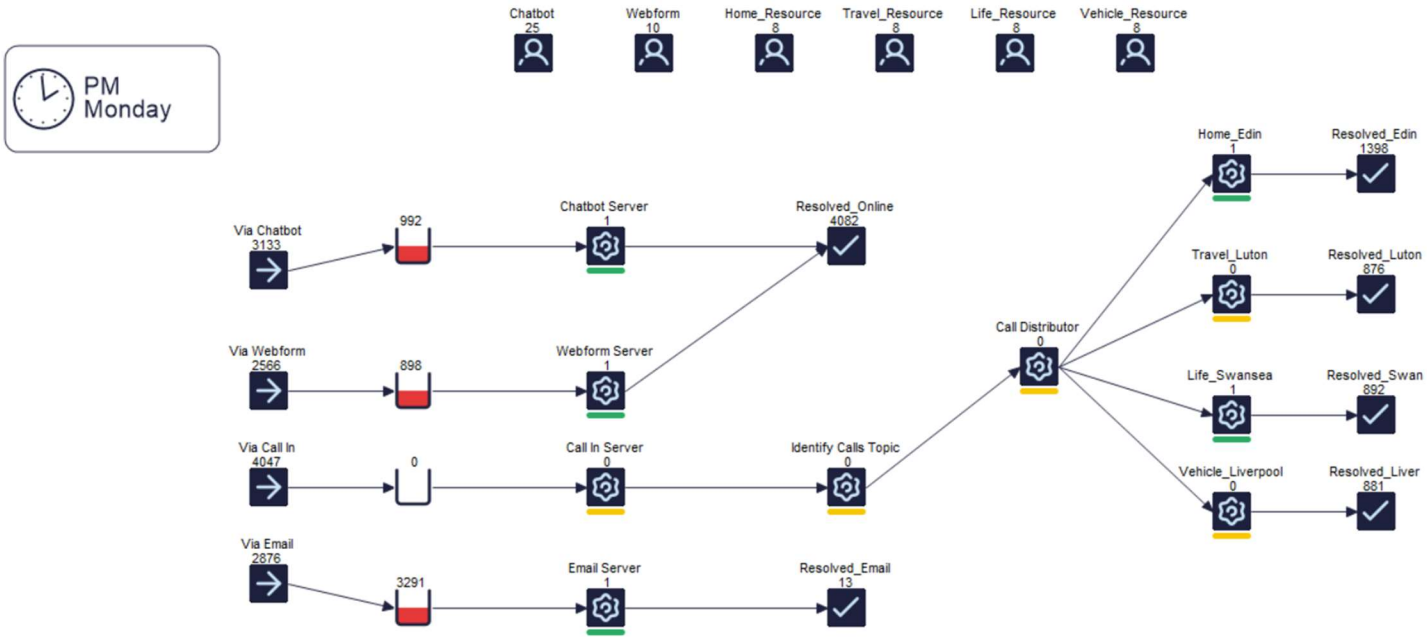


Figure 8: Basic Simul for Current Business Model

7.2.3 Validation of Current Model

The following image shows the key performance indicators with average values and 95% confidence intervals. This system can handle large volumes effectively and hence best to reduce human intervention.

SIMUL8 Results Manager

KPIs | KPI History | Experiments | All Object Results | Custom Reports

		Low 95% Range	Average Result	High 95% Range	Risk
Via Chatbot	Number Entered	3114.51	3152.20	3189.89	
Via Webform	Number Entered	2482.44	2528.60	2574.76	
Via Call In	Number Entered	3958.25	4045.20	4132.15	
Via Email	Number Entered	2807.31	2900.60	2993.89	
Home_Edin	Number Completed Jobs	1343.03	1359.20	1375.37	
Travel_Luton	Number Completed Jobs	872.20	895.60	919.00	
Life_Swansea	Number Completed Jobs	866.96	893.00	919.04	
Vehicle_Liverpool	Number Completed Jobs	868.92	897.00	925.08	
Resolved_Online	Number Completed	4027.71	4066.20	4104.69	
Resolved_Email	Number Completed	13.00	13.00	13.00	

Figure 9: KPI Analysis

8 Management Options

8.1 Option 1

The first option shows the extended AI-assisted technology with 40 chatbots and 20 webforms… with resolving 36 queries doubled than the presented in option 2.

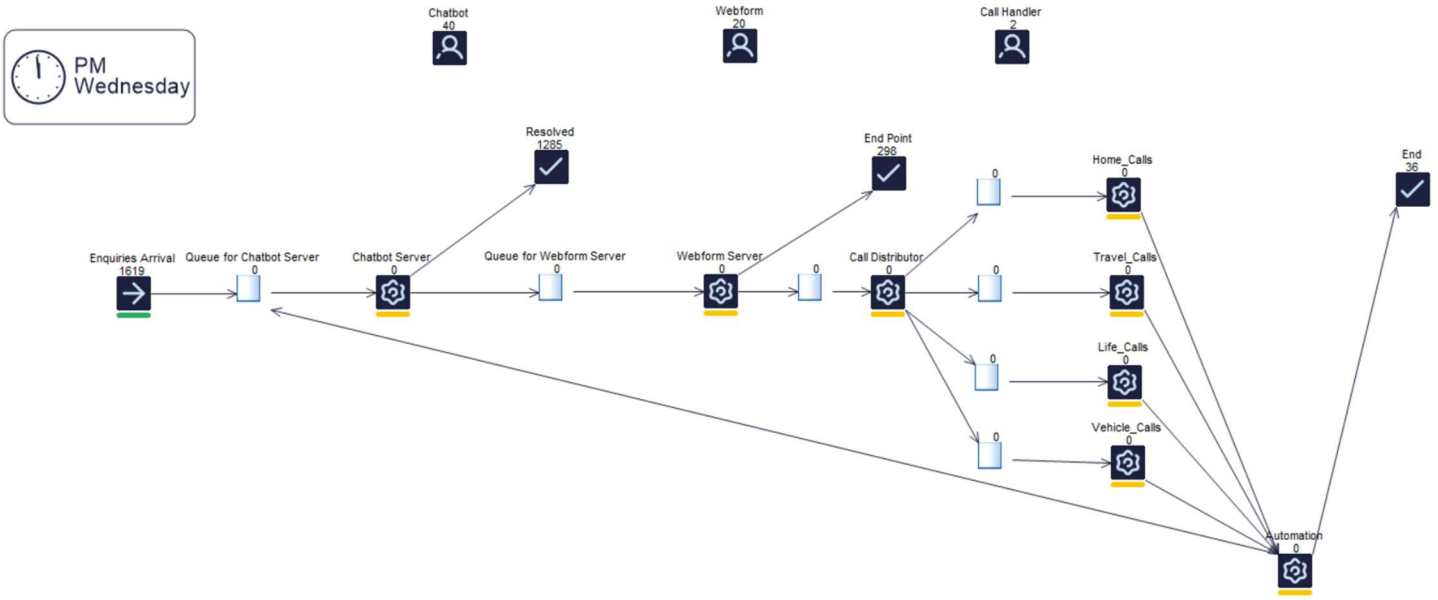


Figure 10: Extended CEOs model 1

8.2 Option 2

This model uses half of the resources so could resolve no more than 551 queries via Chatbot and 130 via Web.

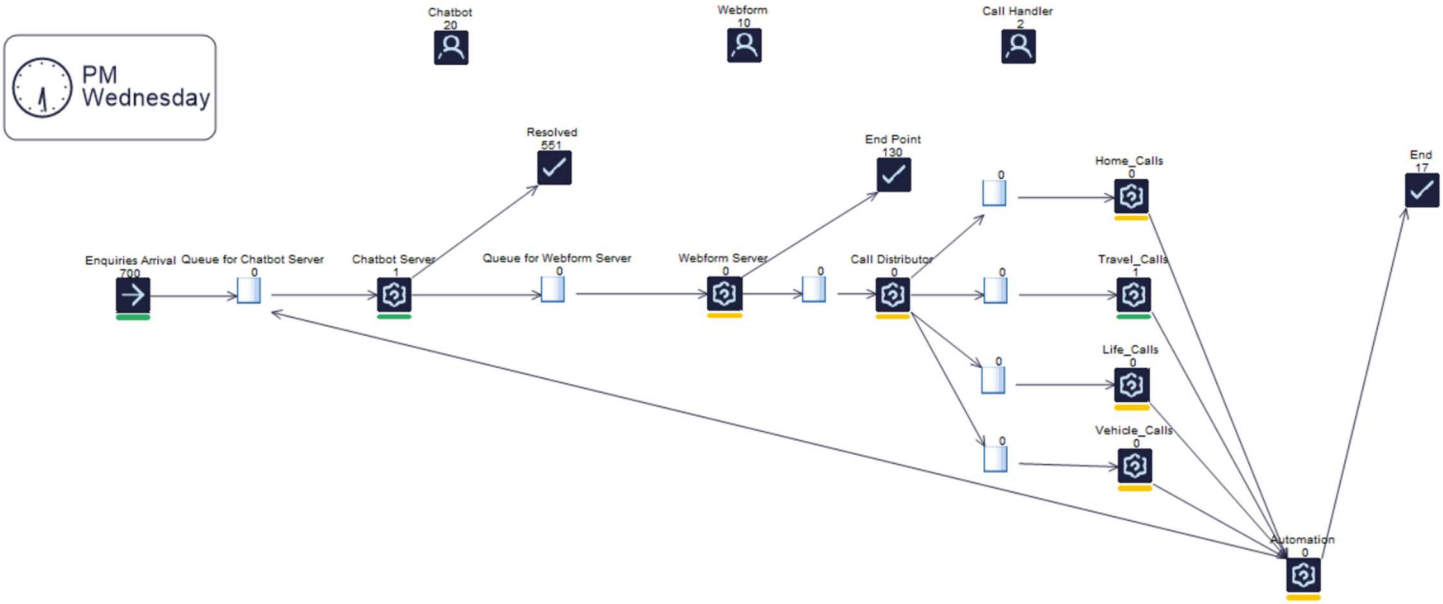


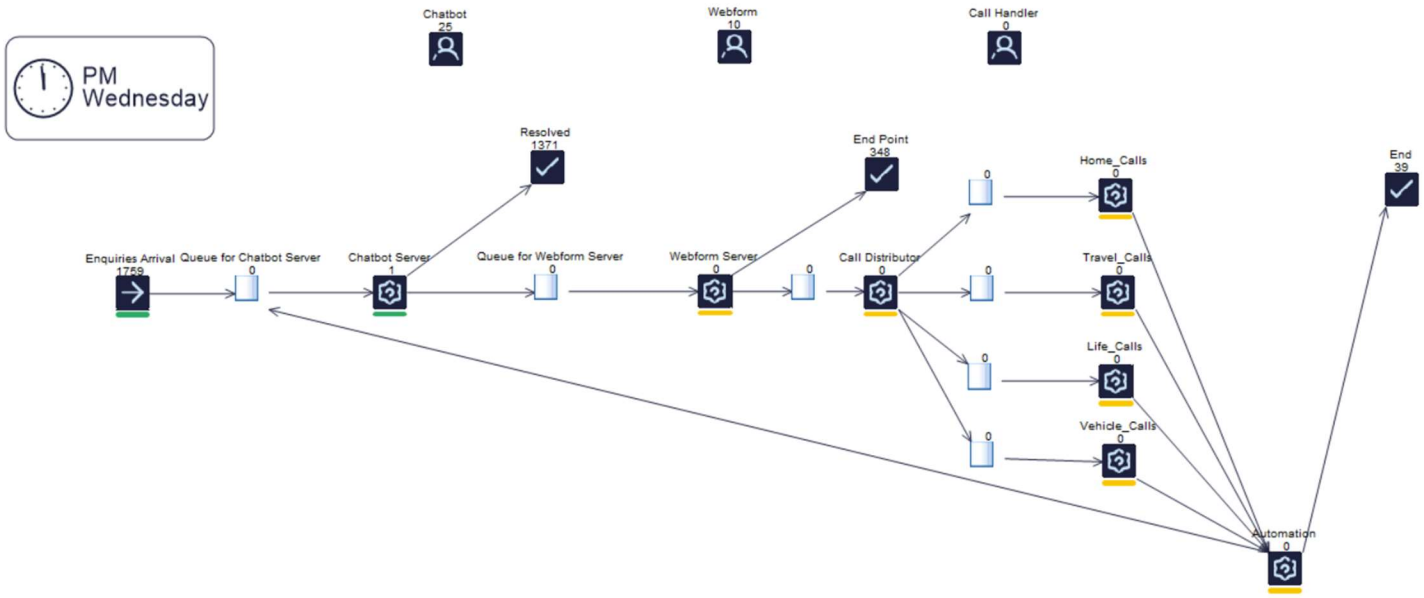
Figure 11: Extended CEOs model 2

8.3 Evaluation of Options

For both the options, it can be drawn that most of the issue types are resolved via chatbots, the remaining ones to webforms, and lastly, enquiries that are not resolved online are routed through the call distributor to Home, Travel, Life, and Vehicle streams. So, it can be concluded that an increased number of chatbots and webforms can resolve customer issues much more effectively and hence reduce dependency on staff, cut budgets, and increase both services and revenue. This also means that most customer issues can be resolved by chatbots and webforms and hence reduce overload from useless enquiries to the human staff. This means investing in AI technology will give the company higher returns compared to increasing human staff.

9 Recommendations For Management

Harbour Insurance Company should invest in automating Large Language Models and AI-assisted technology with 24/7 availability. Based on the requirements to handle large volume of calls and meet SLA standards, I would recommend option 1 to the management team, as it will reduce workload to the staff by automating most of the customer issues. Chatbots should train to resolve billing and payment related issues while human calls should only be opted for when they are in high demand or need serious issues to be resolved. SLAs defined in 2015 should be updated to reflect current demand. Investment should prioritize AI technology over additional staffing to ensure scalability and cost efficiency.

10 Appendices

10.1 Appendix A. CEOs model